
Arthur Petroff

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IT professional and full-stack software developer, dedicated to speed, efficiency, ergonomics, and reliability. Background in STEM tutoring and personal computer/network setup, administration, support, and maintenance.

Skills

7+ years building, installing, managing, and operating computer systems, across Windows and Linux. Thorough technical education in computer and networking systems. 2+ years of independent full-stack web development experience. Strong software and hardware troubleshooting skills.

Skilled with C, HTML/CSS, JavaScript/TypeScript, Rust, Python, C#, Java, Shell Scripting, and SQL. Experienced developing and consuming REST APIs. Expert on the command-line. Experienced with frameworks and toolkits such as React, Bootstrap, Node, and the .NET ecosystem. Fundamental networking skills across TCP/IP and HTTP.

Experienced with tooling such as Git, Hyper-V, Docker, Wireguard, Pangolin, Traefik, and the Microsoft Office suite.

Fluent in English + Brazilian Portuguese.

Education

University of South Carolina Upstate - Bachelor's Degree in Computer Science - 2025

- Notable Coursework: Computer Science I/II/III, C++ Programming, Computer Architecture, Programming Language Structures, Operating Systems, Web Application Development, Software Engineering, Advanced Networking, Network Security, Design & Analysis of Algorithms, Database System Design.
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Work Experience

Full Stack Developer - Self Employment - MAY 2025 to CURRENT

- Developing a full-stack web application to serve third-party game worldstate information to players. Researching back-end architecture optimization.

Student Tutor, Computer Science - USC Upstate - SEP 2024 to MAY 2025

- Assisted students in building mental models of fundamental concepts, encouraging calm, deliberate, and piecemeal problem solving.

Online Instructor - SigmaTeaches - SEP 2021 to JUN 2022

- Led one-on-one virtual tutoring sessions, growing STEM engagement with students ages 6-17.
- Wrote and followed lesson plans for self- or instructor-taught lessons, teaching various computer programming concepts in LUA, through the medium of "ComputerCraft", a programmable computer/robot mod for the video game Minecraft.

- Worked with students in study sessions for “Blender”, a 3D modeling program, developing skills in geometry and physical properties, while interfacing with a modern digital art workflow.
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Projects & Involvement

HackGreenville Community - Greenville, SC

- Frequent contributions to local interest meets, such as the Upstate Carolina Linux Users Group (UCLUG), through event participation and live presentation.

Class Projects - USC Upstate

- Web Framework Comparison Project - Automated HTTP benchmarking and data processing tools to compare the performance of four different modern back-end web server frameworks (ASP.NET, Axum, Express, and Gin), across varying levels of computational stress.
- Budget App Project - Implemented a fully custom multi-threaded web server written in Rust, including a custom endpoint router and parent-child threading architecture enabling per-user thread isolation. Live interactive demonstration hosted on-site on a Raspberry Pi.

Officer of Coding - Computer Science Club, USC Upstate - AUG 2023 to MAY 2025

- Led biweekly interest meets for students on campus, focusing on topics such as competitive coding, game development, back-end web development, and Linux.

Personal Projects

- Portfolio - A portfolio website built utilizing modern frameworks and tooling. Containerized, automated, and self-hosted.
 - nlbt (nice little budget tool) - A command-line budget tracking utility, built for personal use to track expenses.
 - Installed and currently maintaining a private homelab network, including VPN access, off-site reverse proxy IP masking, and a local server.
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References

Available upon request.
